

4 Let $u_r = e^{rx}(e^{2x} - 2e^x + 1)$.

- (a) Using the method of differences, or otherwise, find $\sum_{r=1}^n u_r$ in terms of n and x . [3]

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- (b) Deduce the set of non-zero values of x for which the infinite series

$$u_1 + u_2 + u_3 + \dots$$

is convergent and give the sum to infinity when this exists. [3]

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